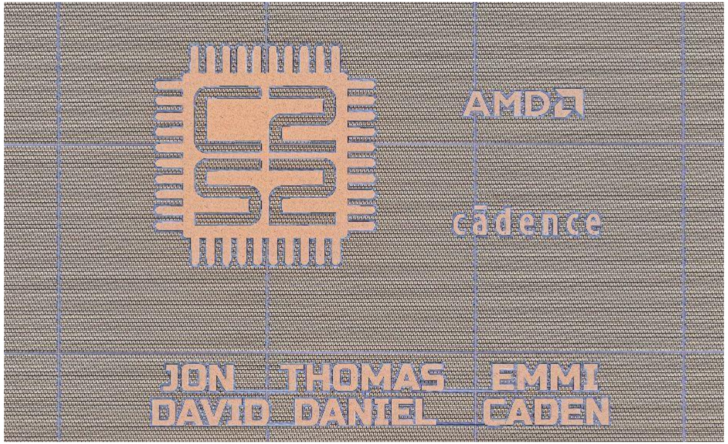
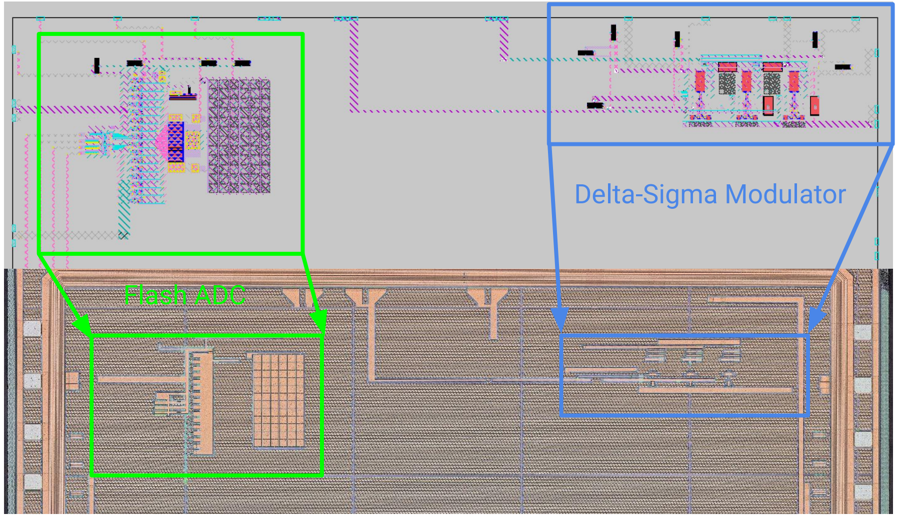
Analog Design

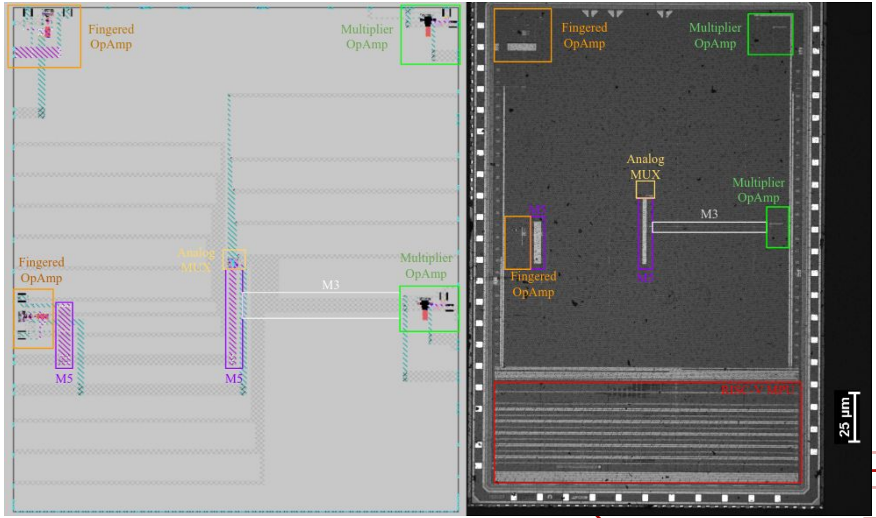
Caden Xu • Daniel Kaminski • Annalise Thompson • Lena Conde Araujo

Past accomplishments

- FA23-SP24 tapeout (Kiwi)
  - Delta-Sigma modulator & Flash-ADC
- FA22-SP23 tapeout (Sparrow)
  - 2 stage operational amplifier

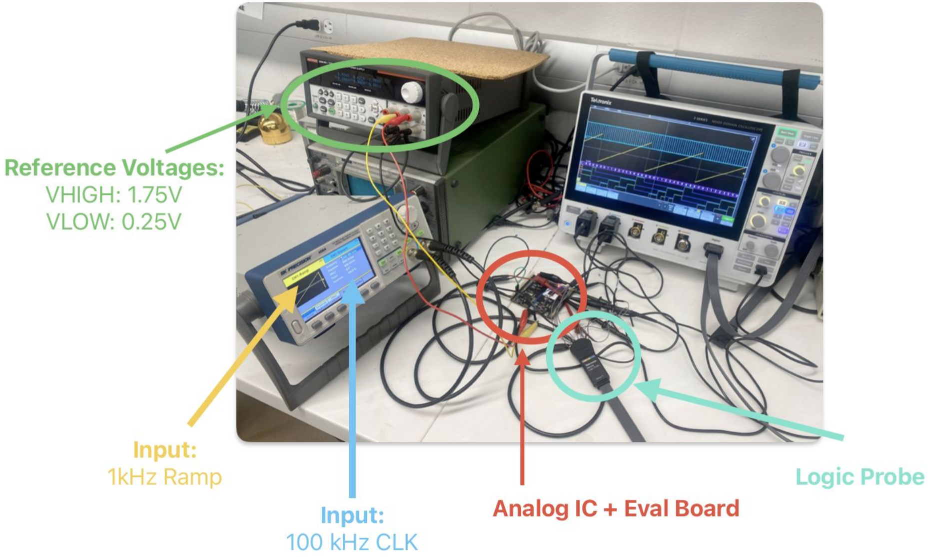






FA24–SP25 Kiwi Flash ADC Testing

- Design spec: 4 bit 40MS/s ADC
- ENOB not good: ~3.5 bits
  - Missing codes!
- Comparator mismatch
- Provides useful **design insight**
  - Inform current year layout+design
- **Monte-Carlo testing** – Random offset mitigation
- **PEX extraction** – Systematic offset mitigation

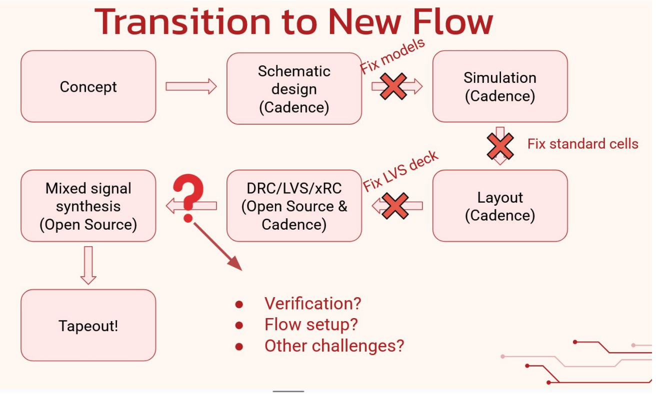


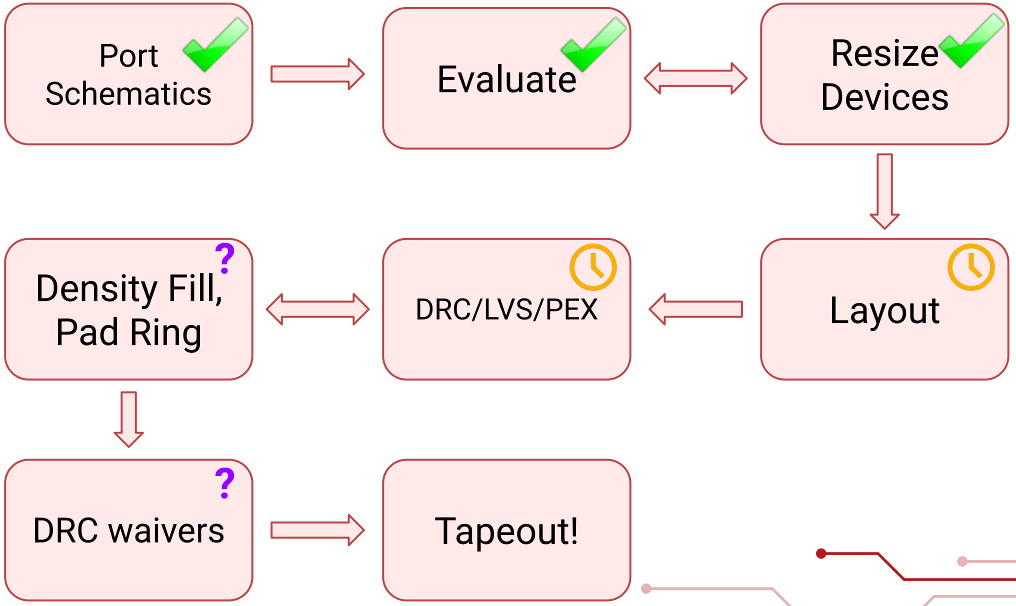
Flow Transition (twice)

A few months ago...
(Open-source to Cadence)

Efabless

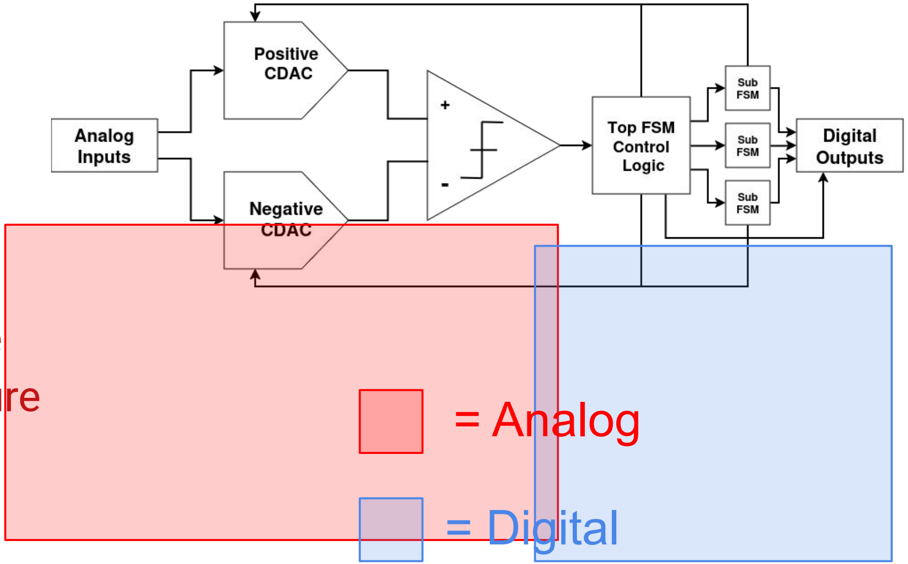
Now... (TSMC Pivot)



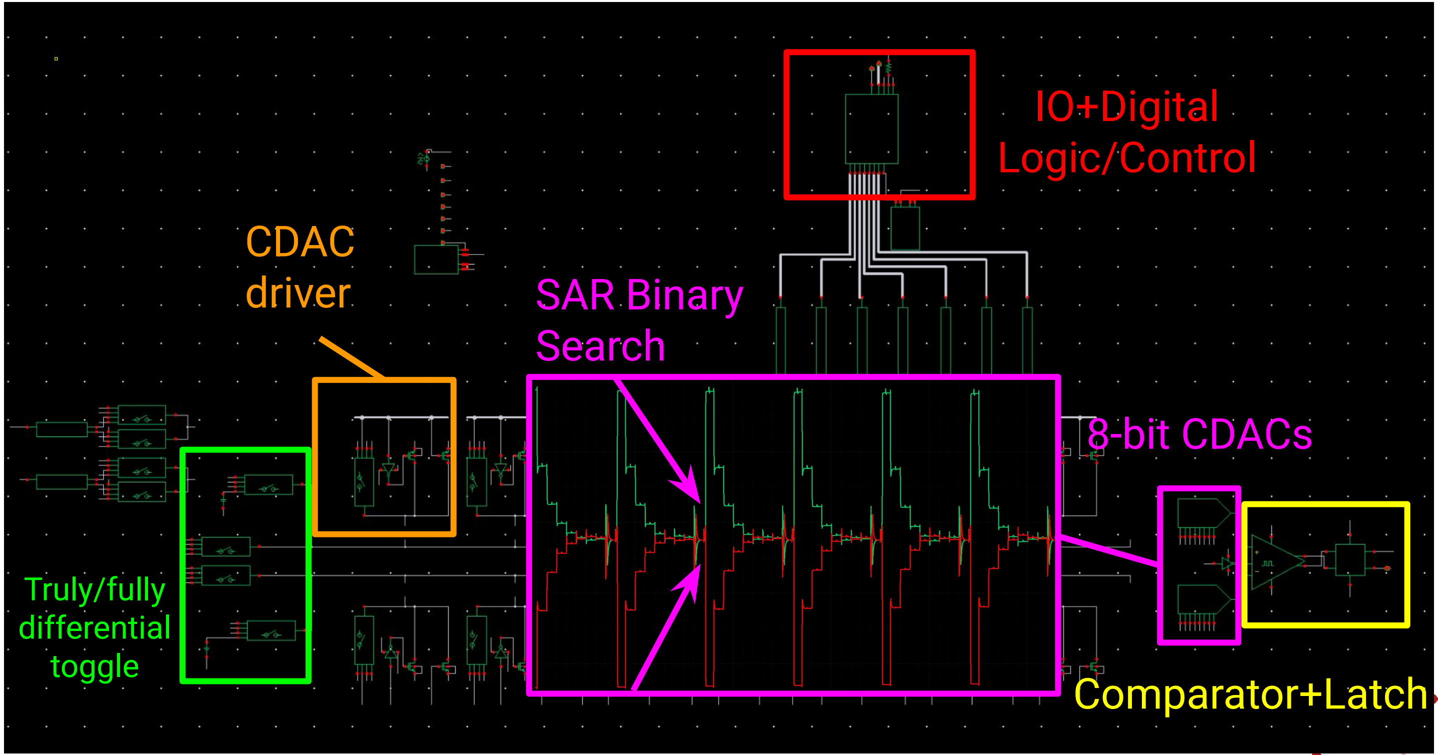


ADC Design Rationale & Architecture

- **High Level Goal:** Sample audio frequency bird chirps for classification and further processing
  - Low power
  - Moderate-high resolution
  - Low sample rate
- Design Choices
  - 8 bit, ~4Msps SAR ADC
  - Fully/Truly differential – togglable
  - Synchronous switching architecture
  - Binary-weighted CDAC
  - Triple-tail clocked comparator
  - Compensated analog switches

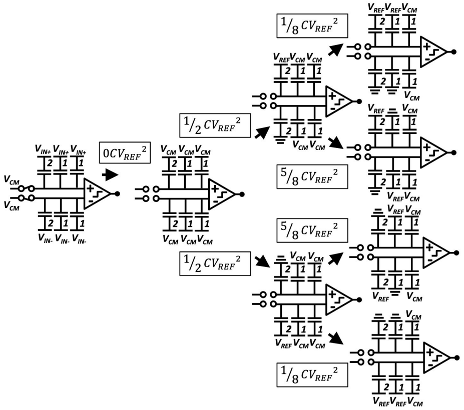


ADC Architecture



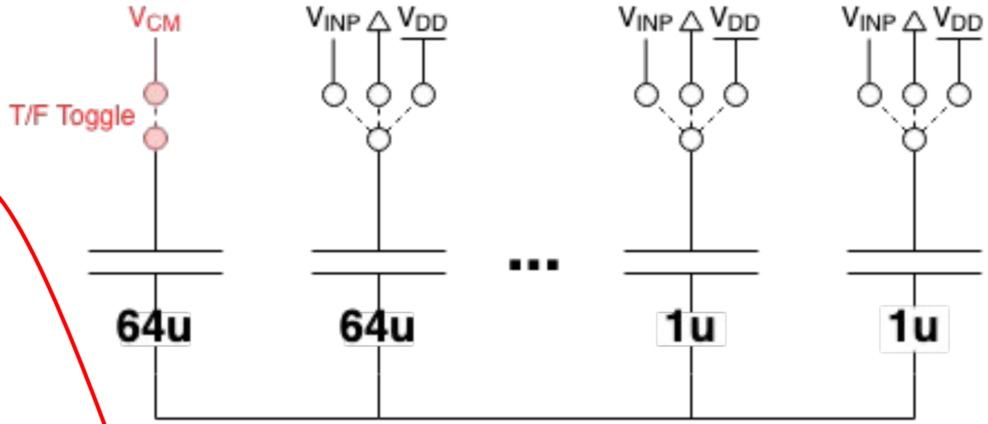
CDAC switching algorithm

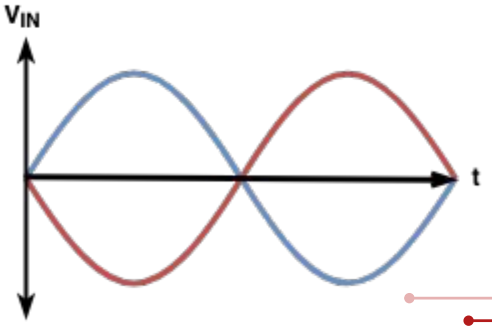
- Inverted merged capacitor sampling
  - Benefits
    - Reduced capacitors $1+2^7$ (rather than $1+2^8$)
    - Bottom plate sampling maintained
    - Lower power
    - Common mode maintained for comparator
  - Drawbacks
    - More complicated algorithm
    - Requires V_{CM} (although high tolerance)
- Major redesign of digital logic code:
  - Incorporating all IO into digital logic
  - Removing unsynthesizable verilog



Truly & Fully Differential Toggle

- In fully differential mode, inputs must maintain a V_{CM}
  - $V_{CM}=½V_{DD}=V_{INP}+V_{INN}$
  - Violation will result in exceeding V_{DD} or GND
- In truly differential mode, inputs can be entirely uncorrelated
- Able to toggle between F/T differential modes
  - Requires an extra 64u capacitor, and an extra analog switch





Example Fully Differential Compliant Input

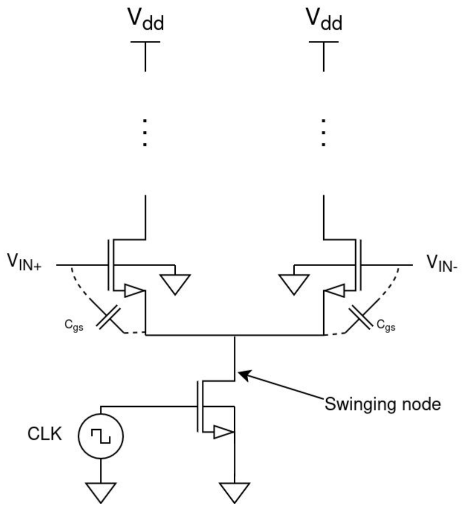
Tape-In: Triple Tail Comparator

- Rationale:
  - Relatively small, but interesting block to tape-out
  - Primary comparator used in SAR ADC
- Challenges:
  - Goal: Offset < 1 LSB/7mV
    - Monte Carlo Simulation
    - Careful layout
    - PEX extraction

    Random offset

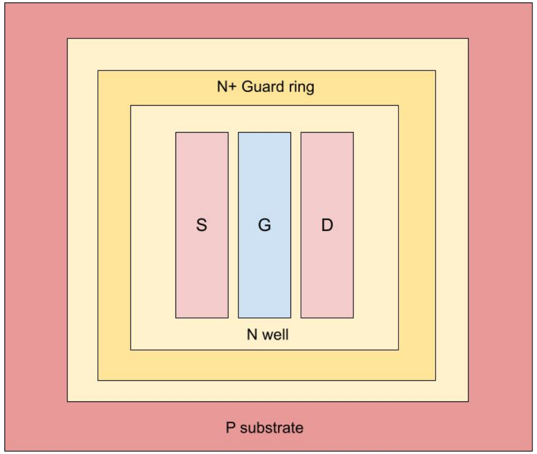
    layout/systematic offset
  - Common mode kickback
    - Tradeoff with offset: Larger input pair
    - Kickback compensation
  - Power draw
- Designed in Fall-24, but ported, modified and laid out

Typical input pair



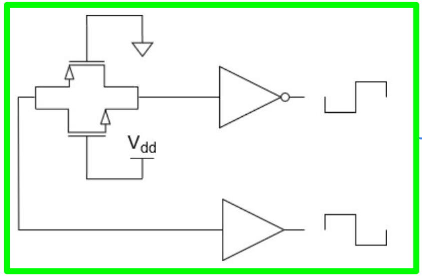
Comparator Layout Approach + Techniques

- Multiple iterations of kickback cancellation
  - PFET dummy devices → NFET dummies + Clock Slew Rate Limiting RC Circuit
  - Dummies reduce ∫ kickback, slew rate limiting reduces max kickback
- Interdigitate fingered devices
- Surround each critical device pair with an associated guard ring (N-Well or P-Tap)
- Advised by ADI engineers to keep orientational symmetry

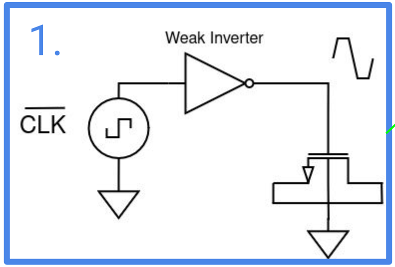


$$_DA_SB_DB_SA_D$$
$$_DB_SA_DA_SB_D$$
$$_DA_SB_DB_SA_D$$
$$_DB_SA_DA_SB_D$$

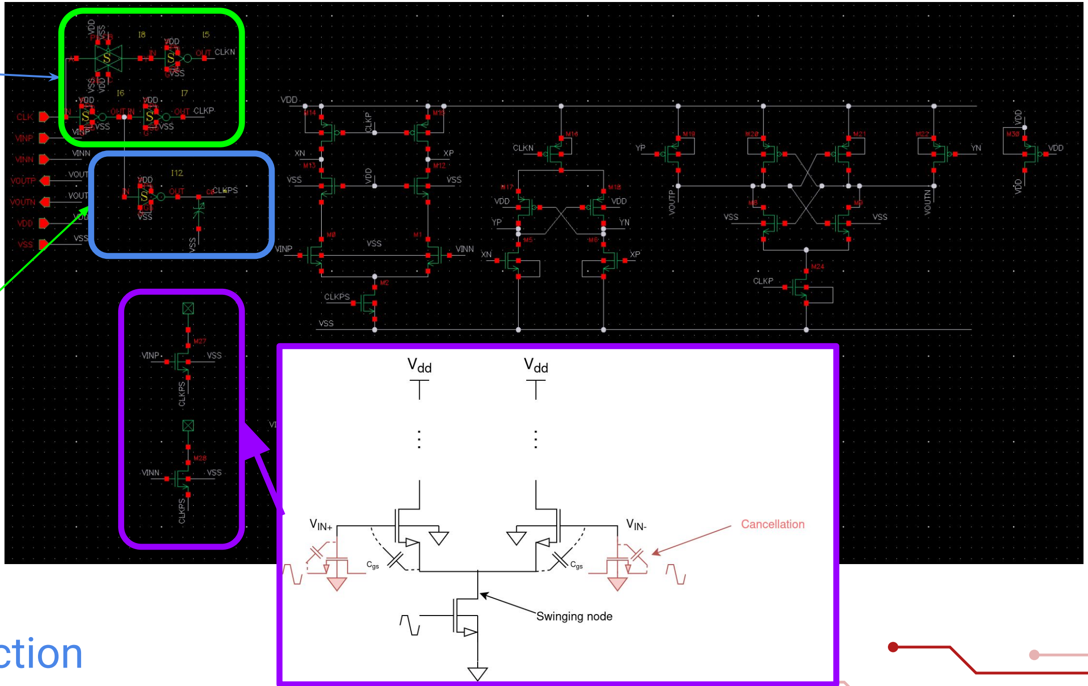
Kickback Compensation



Delay-Matched CLK gen



Slew Rate Control

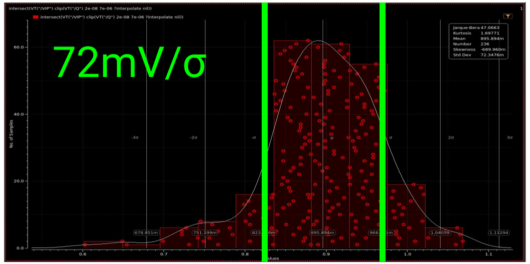


Compensation Pair

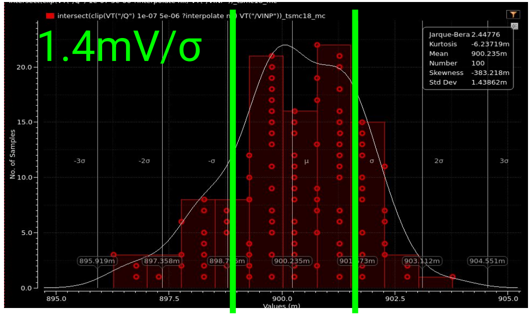
1. Peak/Absolute kickback reduction
2. Integrated kickback reduction

Results

Kiwi Comparator: Previous design

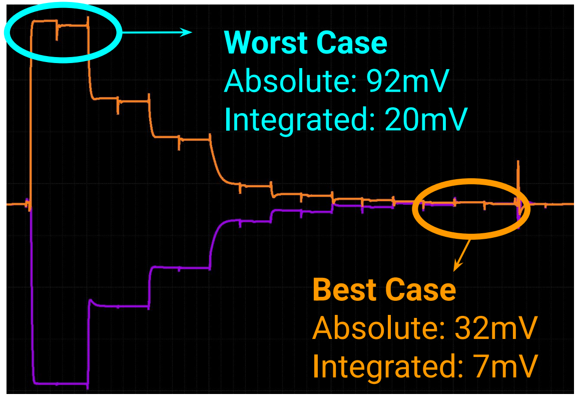


New Design

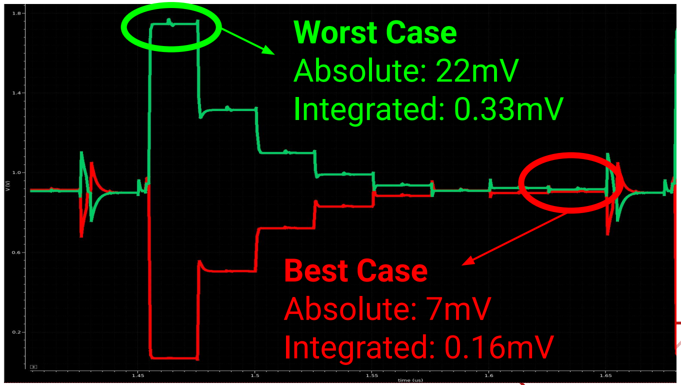


~50x Improvement!

Without compensation



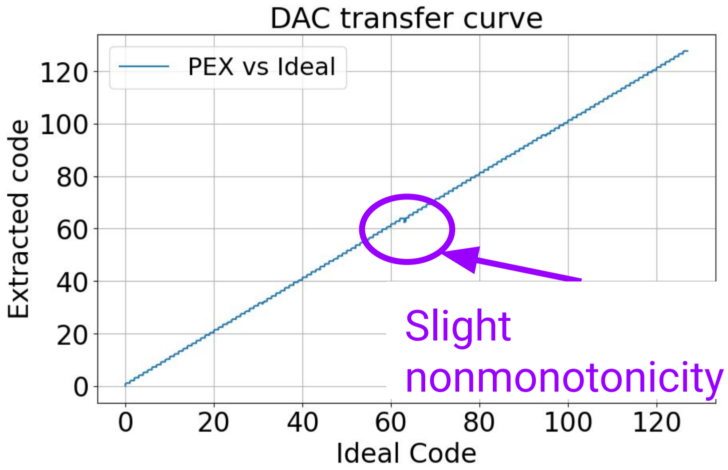
With compensation

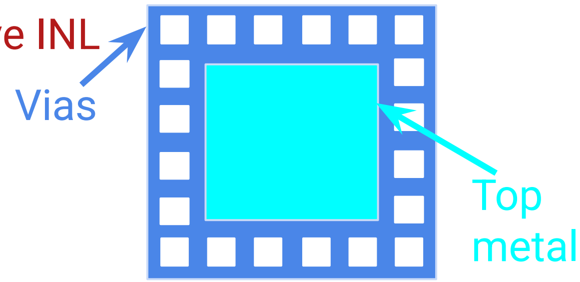


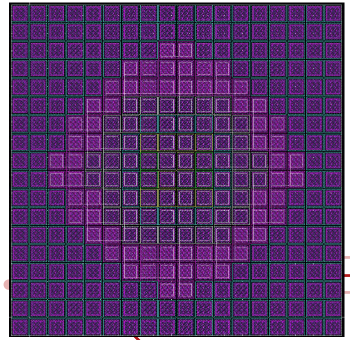
~5x Improvement!

CDAC

- Minimize **random errors**
- Surrounded by 2-thick layer of **dummy devices**
- Routing done on M2-4 between caps
- Many Challenges:
    - **MiM between M5+M6**
    - Metal Density
    - **Antenna rules**
- Will iterate on CDAC to improve INL

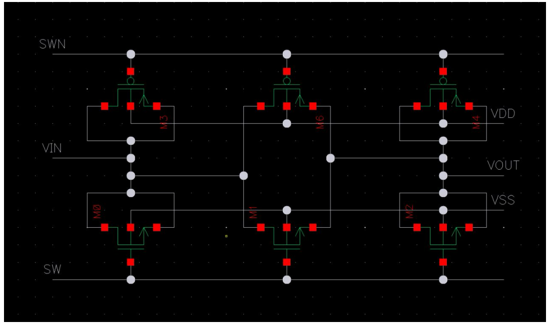




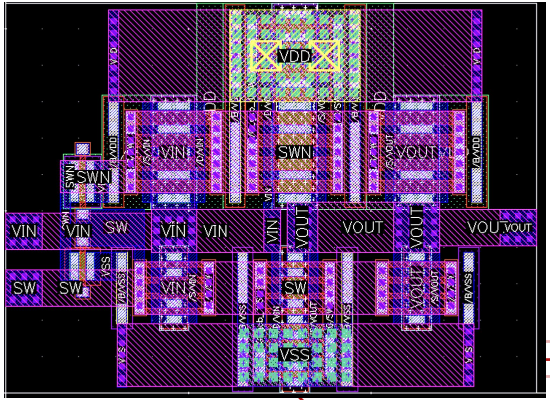


Analog Switches

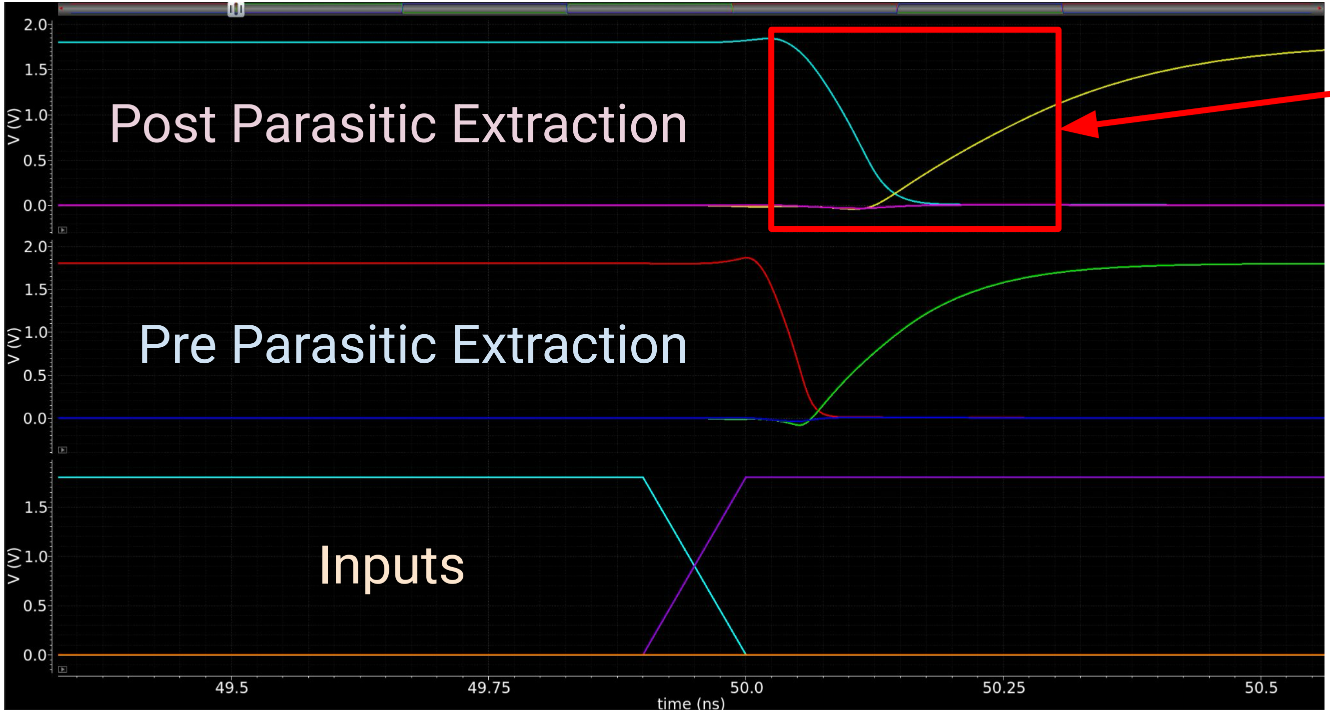
- Ported and optimized **$R_{DS(ON)}$ linearity** and **minimize Q_{inj}**
- **Worst case capacitance to charge:** MSB capacitor
  - 64*U = 64*~31fF = 2pF
  - τ = RC = 2pF * R_{DSon}
- Time constraint: 0.5*(1/40MHz)
  - **On-time is only ½ cycle** or 12.5ns
- Minimum LSB resolution: $1.8/2^8$ = 7mV
- Determining R from RC equation and adding margin
  - $R_{DS(ON)}$ ≈1.01kOhm
- Initially planned bootstrapped switches, instead size for worst case and **don't have to use >1.8V devices**



“Practice” SKY130 layout



Break Before Make Modules



1. Both Below ½ VDD
   ↑ maximize
2. Both Above 10mV
   ↓ minimize

Top Level Simulation Results

- Able to **successfully converge** to digital value with 40MHz clock
    - Including with gate-level simulation of FSM logic
- Still have to complete **top level layout** before post-extraction simulation results are available
- Will continue iterating on CDAC to reduce capacitor mismatch, same for comparator (perhaps sizing smaller for lower power draw)